

Energy of an Undamped Oscillator

The total energy ($E = T + V$) for an undamped oscillator is given by

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2,$$

where $x = A \sin(\omega t + \beta)$ and $\dot{x} = A\omega \cos(\omega t + \beta)$. Therefore,

$$\begin{aligned} E &= \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2, \\ &= \frac{1}{2}mA^2\omega^2 \cos^2(\omega t + \beta) + \frac{1}{2}kA^2 \sin^2(\omega t + \beta), \end{aligned}$$

but $\omega^2 = k/m$, so

$$\begin{aligned} E &= \frac{1}{2}kA^2(\cos^2(\omega t + \beta) + \sin^2(\omega t + \beta)), \\ &= \frac{1}{2}kA^2 = \frac{1}{2}m\omega^2 A^2. \end{aligned}$$

Thus the total energy of a harmonic oscillator is proportional to the amplitude squared.

Worked Example 11.2

A mass of 0.25 kg is attached to a spring of force constant 1.0 N/m. The mass is displaced 0.15 m from its equilibrium point and released with zero initial velocity. Evaluate the total energy of the oscillator (kinetic plus potential). What is the maximum velocity? What is the period of the oscillation?

Solution The total energy is constant. The initial energy is $E = T + V = \frac{1}{2}mv_0^2 + \frac{1}{2}kx_0^2$. But $v_0 = 0$.
 $\therefore E = \frac{1}{2}kx_0^2 = \frac{1}{2}(1)(0.15)^2 = 1.13 \times 10^{-2} \text{ J}.$

The maximum velocity occurs when $x = 0$, so $E = \frac{1}{2}mv_{\max}^2$. Consequently,

$$v_{\max}^2 = \frac{2E}{m} = \frac{(2)(1.13 \times 10^{-2})}{0.25},$$

and $v_{\max} = 0.3 \text{ m/s}.$

Since $\omega = \sqrt{k/m} = \sqrt{1/0.25} = 2$, the period is $P = 2\pi/\omega = \pi$.

Exercise 11.10

Show that $x = x_0 \sin(\omega t + \beta)$ is equivalent to $x = C_1 \cos \omega t + C_2 \sin \omega t$. ■

Exercise 11.11

A spring of constant 10 N/m is connected to a mass of 2 kg. It is initially at rest in an unstretched position. It is given an initial speed of 3 m/s. What is the amplitude of the oscillation? Answer: 1.34 m. ■

11.3 The Damped Harmonic Oscillator

Consider now an oscillator with a retarding force proportional to the velocity. This is a *damped* harmonic oscillator. See [Figure 11.4](#). Common sense tells us that such an oscillator will lose energy and eventually stop.

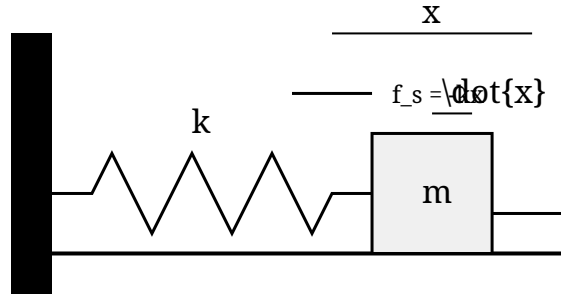


Figure 11.4 A damped harmonic oscillator. At the moment illustrated, the block is moving to the left, toward its equilibrium position. The spring force (f_s) is directed toward the left and the retarding force (f_f) is directed toward the right, opposite to the velocity.

The equation of motion of a damped harmonic oscillator is given by [Equation \(11.6\)](#) which I repeat here:

$$m\ddot{x} + b\dot{x} + kx = 0.$$

This is a second-order linear ordinary homogeneous differential equation with constant coefficients. As previously described, to solve it you make the substitution $x = e^{pt}$, giving

$$mp^2e^{pt} + bpe^{pt} + ke^{pt} = 0.$$

Dividing by e^{pt} yields the auxiliary equation:

$$p^2 + \frac{b}{m}p + \frac{k}{m} = 0.$$

This is a quadratic equation with two solutions:

$$p = -\frac{b}{2m} \pm \sqrt{\left(\frac{b}{2m}\right)^2 - \frac{k}{m}}. \quad (11.11)$$

Let us call the two solutions p_1 and p_2 . Since e^{p_1t} and e^{p_2t} are both solutions to the differential equation, the general solution is the sum of these:

$$x(t) = C_1e^{p_1t} + C_2e^{p_2t}, \quad (11.12)$$

where C_1 and C_2 are constants to be determined from the initial conditions.

If $\left(\frac{b}{2m}\right)^2 > \frac{k}{m}$, the quantity under the square root in [Equation \(11.11\)](#) is positive and the root is

real. Note that the root will be smaller than $\frac{b}{2m}$ so p_1 and p_2 will be negative. Then, according to Equation (11.12), the quantity $x(t)$ will be exponentially decreasing.

On the other hand, if $(\frac{b}{2m})^2 < \frac{k}{m}$ then the quantity under the square root is negative and p_1 and p_2 are complex. That is, they have an imaginary component (call it $i\omega$) and the solution contains terms of the form

$$e^{i\omega t} = \cos \omega t + i \sin \omega t.$$

We have reached an important conclusion. Let me repeat myself for emphasis. From the equation for p , Equation (11.11), you note that p is complex if and only if the quantity under the square root is negative, that is, iff

11.3 THE DAMPED HARMONIC OSCILLATOR

293

$$\left(\frac{b}{2m}\right)^2 < \frac{k}{m}.$$

In this case the system oscillates. On the other hand, if

$$\left(\frac{b}{2m}\right)^2 \geq \frac{k}{m},$$

then p is real and the motion dies out exponentially.

The relation of k/m to $(b/2m)^2$ characterizes the three types of motion for the damped oscillator. These are called "underdamped," "overdamped," and "critically damped." They are described in the following table.

Name	Condition	Description
underdamped	$(b/2m)^2 < k/m$	oscillatory motion
critically damped	$(b/2m)^2 = k/m$	exponential decrease
overdamped	$(b/2m)^2 > k/m$	exponential decrease

The motion of the damped oscillator depends on the relative values of the parameters b, k, m . If $b/2m$ is large, the system is subjected to a considerable retarding force and the oscillations will die out quickly. On the other hand, if $b/2m$ is small, there is little damping and the oscillator

will remain in motion for a long time. Furthermore, if k/m is large, the frequency of oscillation will be large, but if k/m is small, the system will oscillate slowly.

11.3.1 The Underdamped Oscillator

Let us begin by considering the underdamped oscillator. Then, $(b/2m)^2 < k/m$ and Equation (11.11) shows that p will be a complex quantity. It is convenient to change notation at this point. Because $\sqrt{k/m}$ is the "natural" frequency of the undamped oscillator, we shall denote k/m by ω_0^2 . Furthermore, $b/2m$ can be denoted by γ the "damping coefficient." The new notation is :

$$\begin{aligned}\frac{k}{m} = \omega_0^2 &= \text{square of frequency of undamped oscillator,} \\ \frac{b}{2m} = \gamma &= \text{damping coefficient.}\end{aligned}$$

Using this notation, Equation (11.11) becomes

$$p = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}. \quad (11.13)$$

An underdamped oscillator has $\gamma^2 < \omega_0^2$ so the last term is the square root of a negative number, and p is complex. In that case it is convenient to write

$$\sqrt{(-1)(\omega_0^2 - \gamma^2)} = i\omega_1,$$

where $\omega_1 = \sqrt{\omega_0^2 - \gamma^2}$ is a real number. Then

$$p = -\gamma \pm i\omega_1,$$

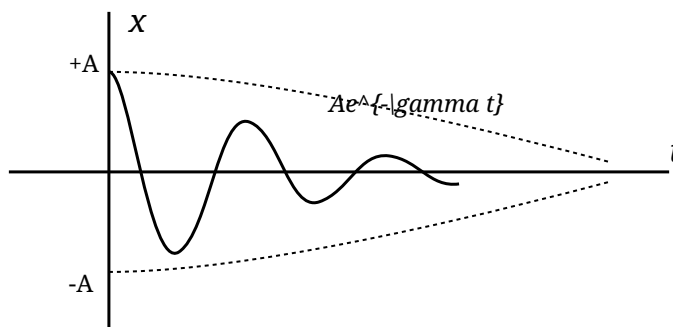


Figure 11.5 The motion of an underdamped oscillator consists of rapid oscillations (described by $\cos(\omega_1 t + \theta)$) modulated by an exponentially decreasing amplitude (described by $Ae^{-\gamma t}$).

a form that clearly shows that p is complex. Inserting the two values of p into the solution (11.12) gives

$$\begin{aligned} x(t) &= C_1 e^{(-\gamma + i\omega_1)t} + C_2 e^{(-\gamma - i\omega_1)t} \\ &= e^{-\gamma t} (C_1 e^{i\omega_1 t} + C_2 e^{-i\omega_1 t}). \end{aligned}$$

C_1 and C_2 are complex constants related to the initial conditions. These constants can be written in a variety of different ways. You obtain a very nice-looking expression if you write them in the form

$$C_1 = \frac{1}{2} A e^{i\theta} \quad \text{and} \quad C_2 = \frac{1}{2} A e^{-i\theta}. \quad (11.14)$$

This basically replaces the two complex constants C_1 and C_2 by the real constants A and θ . Then

$$x(t) = e^{-\gamma t} \frac{1}{2} A (e^{i(\omega_1 t + \theta)} + e^{-i(\omega_1 t + \theta)}),$$

or

$$x(t) = A e^{-\gamma t} \cos(\omega_1 t + \theta). \quad (11.15)$$

Plotting x vs. t generates an oscillatory solution (due to the cosine term), but the amplitude of the oscillations is $A e^{-\gamma t}$, so the amplitude decreases exponentially with time. This behavior is illustrated in [Figure 11.5](#).

Note that the oscillations have an angular frequency ω_1 , which is smaller than the natural frequency ω_0 . That is, the damped system oscillates more slowly than the undamped system. You probably expected this type of behavior.

Worked Example 11.3

An underdamped harmonic oscillator has $k = 2$ N/m, $m = 1$ kg, and $b = 0.1$ kg/s. How many oscillations does the system make before the amplitude decreases to $1/e$ of its initial value?

Solution The time when the amplitude is $1/e$ of its initial value is obtained from

$$Ae^{-\gamma t} = \frac{1}{e}A \quad \therefore t = 1/\gamma = \frac{2m}{b} = 20 \text{ s.}$$

Then,

$$\begin{aligned} \omega_1 t &= \left(\sqrt{\omega_0^2 - \gamma^2} \right) t = \left(\sqrt{k/m - \gamma^2} \right) t \\ &= \left(\sqrt{2 - 0.05^2} \right) (20) = 28.26 \text{ rad,} \end{aligned}$$

and the number of oscillations is $28.26/2\pi = 4.5$.

Exercise 11.12

The complex constants C_1 and C_2 can be written in the form $C_1 = a_1 + ib_1$ and $C_2 = a_2 + ib_2$. But there can only be two independent constants, so a_1, b_1 and a_2, b_2 are not all independent. Using Equation (11.14) show that $a_2 = a_1$ and $b_2 = -b_1$. ■

Exercise 11.13

Obtain the general solution of the following differential equations: (a) $m\ddot{x} + b\dot{x} - kx = 0$, (b) $m\ddot{x} - b\dot{x} + kx = 0$. Assume m, b , and k are positive real quantities. ■

11.3.2 The Overdamped Oscillator

Now consider the *overdamped* oscillator, defined to be an oscillator for which

$$\left(\frac{b}{2m} \right)^2 > \frac{k}{m}.$$

According to Equation (11.11) or Equation (11.13), p will now have two *real* values denoted γ_1 and γ_2 . Using the notation of Equation (11.13), the solutions of the auxiliary equation are

$$p_1 = -\gamma_1 = -\gamma - \sqrt{\gamma^2 - \omega_0^2},$$

and

$$p_2 = -\gamma_2 = -\gamma + \sqrt{\gamma^2 - \omega_0^2}.$$

Note that both γ_1 and γ_2 are positive and that γ_1 is greater than γ_2 .

Plugging these expressions into the general solution of the differential equation (11.12) leads to

$$x(t) = C_1 e^{-\gamma_1 t} + C_2 e^{-\gamma_2 t}, \quad (11.16)$$

where γ_1 and γ_2 are real. Both terms are exponentially decreasing with time, as indicated in Figure 11.6. Because $\gamma_1 > \gamma_2$ we note that $e^{-\gamma_1 t}$ decays more rapidly than $e^{-\gamma_2 t}$.

The motion of the oscillator (that is, the value of x as a function of time) is, of course, the sum of two curves, as illustrated in Figure 11.6. The constants C_1 and C_2 depend on the initial conditions. For example, if you pull the block to some position x_0 and release it from rest ($v_0 = 0$), at $t = 0$, Equation (11.16) reduces to

$$x_0 = C_1 + C_2 \quad \Rightarrow \quad C_1 = x_0 - C_2.$$

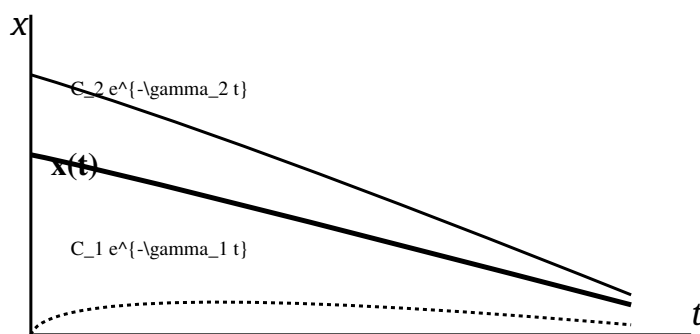


Figure 11.6 An overdamped harmonic oscillator. The motion is the sum of the two curves. The constants C_1 and C_2 depend on the initial conditions.

Since $\dot{x} = -\gamma_1 C_1 e^{-\gamma_1 t} - \gamma_2 C_2 e^{-\gamma_2 t}$,

$$v_0 = -\gamma_1 C_1 - \gamma_2 C_2.$$

But $v_0 = 0$, so $C_2 = -(\gamma_1/\gamma_2)C_1$, and hence,

$$C_1 = x_0 - C_2 = x_0 + (\gamma_1/\gamma_2)C_1.$$

Therefore,

$$C_1 = -\frac{\gamma_2}{\gamma_1 - \gamma_2}x_0,$$

and

$$C_2 = \frac{\gamma_1}{\gamma_1 - \gamma_2}x_0.$$

Consequently,

$$x(t) = \frac{\gamma_2}{\gamma_1 - \gamma_2}x_0e^{-\gamma_1 t} + \frac{\gamma_1}{\gamma_1 - \gamma_2}x_0e^{-\gamma_2 t}.$$

Depending on the initial conditions the oscillator will behave in one of three ways. The displacement $x(t)$ may rapidly decay to zero. Or $x(t)$ may increase for a time and then decay to zero. Finally, $x(t)$ could change sign, reach a local maximum or minimum, then approach zero.

Exercise 11.14

Consider the motion of an overdamped oscillator. Assume that initially $x(t=0) = 1$ and $\dot{x}(t=0) = 0$. Let $\gamma_1 = 3.414$ and $\gamma_2 = 0.586$. Plot $x(t)$. ■

Exercise 11.14

Obtain C_1 and C_2 for an overdamped oscillator if the initial conditions are $x(t=0) = 0$ and $\dot{x}(t=0) = v_0$. Answer: $C_1 = v_0/(\gamma_2 - \gamma_1)$; $C_2 = -C_1$. ■

11.3.3 The Critically Damped Oscillator

Finally, consider the case of a critically damped harmonic oscillator for which

$$\left(\frac{b}{2m}\right)^2 = \frac{k}{m}.$$

Now the term under the root in [Equation \(11.11\)](#) is zero.

For convenience, define the quantity γ_c as follows:

$$\gamma_c = \frac{b}{2m} = \sqrt{\frac{k}{m}} = \omega_0.$$

Then Equation (11.13) yields a single value for p namely $p = -\gamma_c$. Recall that if p has only one value, the solution to a second-order differential equation has the form of Equation (11.9), that is

$$x(t) = C_1 e^{-\gamma_c t} + C_2 t e^{-\gamma_c t},$$

or

$$x(t) = (C_1 + C_2 t) e^{-\gamma_c t}.$$

For critical damping x decays quickly to zero.

Exercise 11.13

A critically damped harmonic oscillator is initially at its equilibrium position with velocity v_0 . Obtain expressions for C_1 and C_2 . Answer: $C_1 = 0$, $C_2 = v_0$. ■

Exercise 11.17

A critically damped harmonic oscillator has $b = 0.3$ kg/s and $k = 0.4$ N/m. (You are not given the mass.) It is released from rest at $x = 0.04$ m. Evaluate the constants C_1 and C_2 . Answer: $C_1 = 0.04$, $C_2 = 0.107$ (in meters). ■

11.4 The Forced Harmonic Oscillator

11.4.1 Statement of the Problem

A *forced* harmonic oscillator is an oscillator which is subjected to some sort of driving force. The external driving force is often periodic with the same frequency as the oscillator. When you push your little brother on a swing, you give the seat a shove each time it comes back to you at the top of its swing. The periodicity of your shoves is the same as the natural period of the swing. Similarly, the pendulum in a "grandfather" clock receives a small push on each swing. An old-fashioned pocket watch has an oscillating spring wheel that receives small periodic impulses

from a spring and ratchet system.

The oscillator will, in general, be damped, so the equation of motion for such a system is

$$m\ddot{x} + b\dot{x} + kx = F(t), \quad (11.17)$$

where $F(t)$ is the driving force. The purpose of the driving force is to overcome the damping. If $F(t)$ is too small, the oscillations will gradually die out. On the other hand, if $F(t)$ is too large, the amplitude may grow unacceptably large. (If you push the swing too hard it will "go over the top," leading to unpleasant consequences.) As you shall see shortly, the amplitude of the oscillations also depends on the *frequency* of the driving force. This is related to the phenomenon of resonance that we will consider later.

The equation of motion of the driven harmonic oscillator Equation (11.17) is an *inhomogeneous* second-order linear differential equation. I will now make a short aside to describe how to solve such an equation.

11.4.2 Inhomogeneous Linear Ordinary Differential Equations

The title of this section describes the kind of equation we want to solve. You should understand the implication of each of the many longish words in the description. The equation is, of course, a *differential equation* because there are derivatives in it. It is *linear* because the dependent variable (x) and its derivatives are not raised to a power larger than the first. We are dealing with an *ordinary* differential equation because Equation (11.17) has no partial derivatives in it.

You probably recall that differential equations are often expressed in terms of the *differential operator* $D \equiv \frac{d}{dt}$. Then $D^2 = \frac{d^2}{dt^2}$ and so on. A linear homogeneous differential equation is an equation which can be put in the form

$$a_n D^n x + a_{n-1} D^{n-1} x + \cdots + a_1 D x + a_0 x = 0.$$

Although the a_i 's are usually constants, they can be functions of t . Note the zero on the right-hand side.

Of course, the equation need not have a zero on the right. There could be a function of x and t on the right-hand side. Thus,

$$a_n D^n x + a_{n-1} D^{n-1} x + \cdots + a_1 D x + a_0 x = f(x, t).$$

Such an equation is said to be *inhomogeneous*. The solution of an inhomogeneous differential equation can be determined using the following rule.

Rule 3. The solution of an *inhomogeneous* differential equation is the sum of the general solution of the corresponding *homogeneous* equation and a *particular solution* of the inhomogeneous equation.

Therefore, the first step in solving an inhomogeneous differential equation is to set the right-hand side equal to zero. This generates the corresponding homogeneous differential equation which we can solve by the method of [Section 11.2](#). The general solution will, of course, contain as many arbitrary constants as the order of the equation. A second-order equation has a general solution with two arbitrary constants. We denote this general solution of the homogeneous equation by x_g . It is usually called the "complementary function." Next we consider the *inhomogeneous* equation. We can sometimes figure out a solution by inspection. We denote this *particular solution* by x_p . The general solution of the inhomogeneous equation is $x_p + x_g$. It is easy to prove this assertion, and it is left as an exercise for you. We now return to the problem of the driven harmonic oscillator.

Exercise 11.18

Show that the general solution of an inhomogeneous differential equation is the sum of a particular solution and the general solution of the homogeneous differential equation. ■

11.4.3 Obtaining the Particular Solution

The equation of motion for the driven, damped harmonic oscillator is

$$m\ddot{x} + b\dot{x} + kx = F(t).$$

The forcing term, $F(t)$, is often periodic because that will keep the oscillator in motion, counteracting the effect of damping. Another common situation is an oscillator that has been subjected to an impulsive force. In general, the forcing term can be any function of time. It could even be a constant. In fact, for the sake of making the analysis simple, I will begin with a constant force. For example, a mass hanging from a spring in an external gravitational field (see [Figure 11.7](#)) has the equation of motion

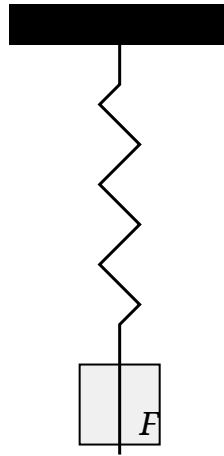


Figure 11.7 A mass hanging from a spring in a constant gravitational field.

$$m\ddot{x} + b\dot{x} + kx = mg, \quad (11.18)$$

where the damping could be due to air resistance, for example. The mass is only allowed to oscillate vertically; it does not simultaneously swing like a pendulum.

The corresponding homogeneous equation was solved in the previous section. If the oscillation is underdamped, for example, the general solution of the homogeneous equation is given by [Equation \(11.15\)](#) as:

$$x_g = x(t) = Ae^{-\gamma t} \cos(\omega_1 t + \theta).$$

Now you need to find a particular solution to [Equation \(11.18\)](#). That is, you need to find a function of x which satisfies (11.18). After a little creative staring at that equation you will realize that if x is constant, the first two terms are zero and you are left with $kx = mg$. Therefore, $x = mg/k$ is a possible solution to the equation. It is the *particular solution*, x_p . The general solution of the inhomogeneous equation is, therefore,

$$x(t) = x_g + x_p = Ae^{-\gamma t} \cos(\omega_1 t + \theta) + \frac{mg}{k}.$$

This tells you that the effect of the gravitational force is just to stretch the spring, giving a different (lower) value for the equilibrium point.

I will now go on to a more interesting problem, namely a driving force that varies sinusoidally.⁴ Let the driving frequency be denoted ω_d and assume $F(t) = F_0 \sin \omega_d t$. For simplicity, we shall begin with an *undamped oscillator*. Then the equation of motion is

$$m\ddot{x} + kx = F_0 \sin \omega_d t,$$

or

$$\ddot{x} + \omega_0^2 x = \frac{F_0}{m} \sin \omega_d t. \quad (11.19)$$

The general solution of the homogeneous equation is

$$x_g(t) = A \cos(\omega_0 t + \theta_0),$$

where A and θ_0 depend on the initial conditions. For example, $\theta_0 = 0$ corresponds to the oscillator at position $x = A$ at time $t = 0$. (For a pendulum this means that at time zero the bob was at the top of the swing, and for a mass on a spring, it means that at time zero the spring was fully extended.)

⁴ Recall that any periodic function can be expressed as a sum of sines and cosines (a "Fourier series") so it is reasonable to consider a sinusoidal driving force.

Having a general solution to the homogeneous equation, you still need a particular solution to the inhomogeneous equation. Previously, we obtained a particular solution by inspection. This is usually possible, so fortunately there are techniques available for determining a particular solution.⁵ One technique that always works for a second-order differential equation is to integrate the equation twice. To apply the method, first write the equation in the form

$$(a_2 D^2 + a_1 D + a_0)x = f(t). \quad (11.20)$$

This equation can be expressed as the product

$$(D - b_1)(D - b_2)x = f(t). \quad (11.21)$$

Let $u = (D - b_2)x$. Then the differential equation is

$$(D - b_1)u = f(t).$$

You now have a first-order linear differential equation that you can solve for $u(t)$. Next, going back to the definition of u , you solve

$$(D - b_2)x(t) = u(t)$$

for $x(t)$. This technique produces the general solution for the inhomogeneous equation, that is, it generates the complementary function as well as the particular solution. However, it may involve a considerable amount of mathematics, so I want to show you two other techniques which might be classified as "tricks" for finding the particular solution. (When a mathematical trick becomes very familiar we quit calling it a trick and start calling it a technique!)

Trick #1

The first trick (or technique) is used when the function on the right-hand side is of the form $f(t) \propto e^{at}$. Then, writing the second-order differential equation in the form

$$(D - b_1)(D - b_2)x = Ke^{at}, \quad (11.22)$$

the particular solution is

$$\begin{aligned} x_p &= Ce^{at} & \text{if } a \neq b_1 \text{ and } a \neq b_2 \text{ and } b_1 \neq b_2, \\ x_p &= Cte^{at} & \text{if } a = b_1 \text{ or } a = b_2 \text{ but } b_1 \neq b_2, \\ x_p &= Ct^2e^{at} & \text{if } a = b_1 = b_2, \end{aligned} \quad (11.23)$$

where C is determined by plugging x_p back into the differential equation.

Trick #2

The second trick is based on the first trick and we use it when $f(t)$ (the right-hand side) is either a sine or a cosine. In that case, write the equation in one of the two following forms:

$$(D - b_1)(D - b_2)\xi_p = \begin{cases} K \sin at \\ K \cos at \end{cases}, \quad (11.24)$$

where ξ_p is a complex function of t . Note that $K \sin at$ is the imaginary part of Ke^{iat} and $K \cos at$ is the real part of Ke^{iat} . We then solve

$$(D - b_1)(D - b_2)\xi_p = Ke^{iat}$$

using formulas (11.23) and take either the real or the imaginary part of the answer. That is, the particular solution x_p is either $\text{Re}(\xi_p)$ or $\text{Im}(\xi_p)$.

5 See, for example, Mary L. Boas, *Mathematical Methods in the Physical Sciences*, 3rd ed., John Wiley & Sons Inc., Hoboken, NJ, 2006, p. 417 ff.